EXPERIMENT 5: Gray-Level Transformations: Negative, Log, Power-Law, Contrast Stretching and Intensity Slicing

1. Aim

To enhance images using point (gray-level) transformations, including image negative, log transformation, power-law (gamma) transformation, contrast stretching and intensity-level slicing using MATLAB.

2. Software Required

- Personal computer
- MATLAB R2020a or later
- Image Processing Toolbox

3. Program

Save the following MATLAB program as exp05.m.

```
clc;

clear;

close all;


%% Read the Original Image

I = imread('pout.tif');

r = im2double(I);


%% Part A: Gray-Level Transformations

s_neg = 1 - r;

s_log = mat2gray(log(1 + r));

s_g04 = r .^ 0.4;

s_g25 = r .^ 2.5;

s_cs = imadjust(I, stretchlim(I), []);


figure('Name','Gray-Level Transformations');
```

```
subplot(2,3,1);
imshow(r);
title('Original');

subplot(2,3,2);
imshow(s_neg);
title('Negative');

subplot(2,3,3);
imshow(s_log);
title('Log Transformation');

subplot(2,3,4);
imshow(s_g04);
title('Gamma = 0.4');

subplot(2,3,5);
imshow(s_g25);
title('Gamma = 2.5');

subplot(2,3,6);
imshow(s_cs);
title('Contrast Stretched');

%% Part B: Transformation Curves
rr = linspace(0, 1, 256);

figure('Name','Transformation Functions');
```

```
plot(rr, 1-rr, ...
    rr, log(1+rr)/log(2), ...
    rr, rr.^0.4, ...
    rr, rr.^2.5, ...
    rr, rr, 'k--');

legend('Negative','Log','Gamma = 0.4', ...
      'Gamma = 2.5','Identity', ...
      'Location','southeast');

xlabel('Input Level r');
ylabel('Output Level s');
grid on;
axis square;

%% Part C: Contrast Stretching of a Colour Image
RGB = imread('peppers.png');

figure('Name','Colour Contrast Stretching');

subplot(1,2,1);
imshow(RGB);
title('Original Image');

subplot(1,2,2);
imshow(imadjust(RGB, [0.2 0.8], []));
title('Contrast Stretched');
```

```
%% Part D: Intensity-Level Slicing
A = 0.40;
B = 0.70;

slice1 = r;
slice1(r >= A & r <= B) = 1;

slice2 = double(r >= A & r <= B);

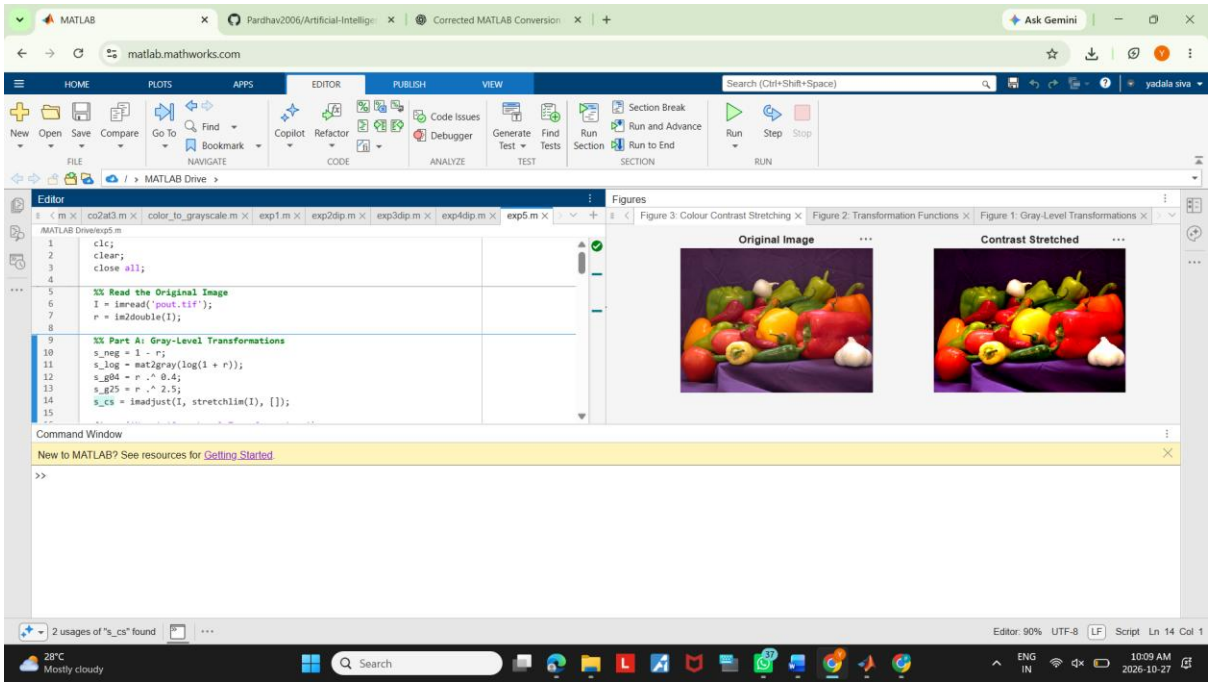
figure('Name','Intensity-Level Slicing');

subplot(1,3,1);
imshow(r);
title('Original');

subplot(1,3,2);
imshow(slice1);
title('Band Highlighted');

subplot(1,3,3);
imshow(slice2);
```

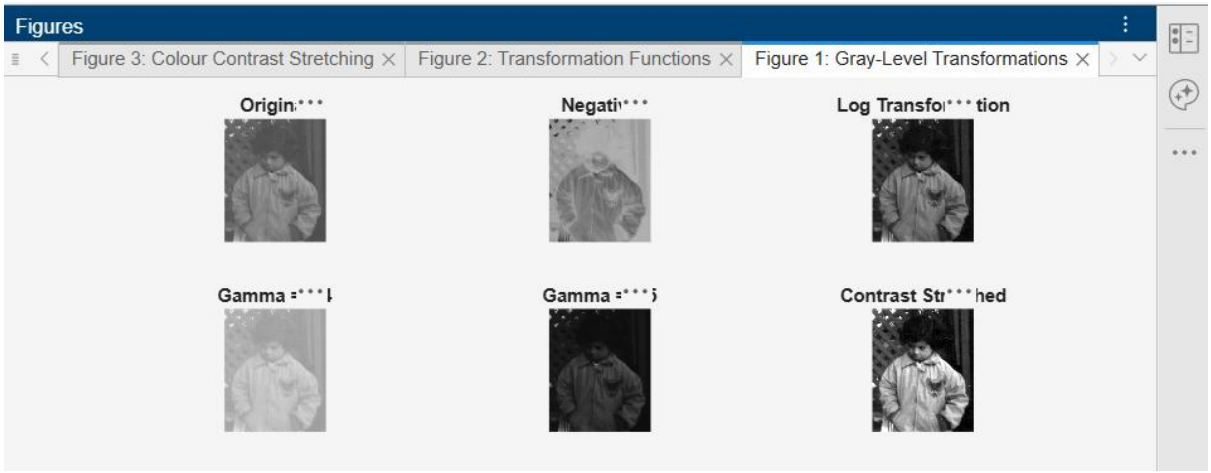
title('Binary Slice');

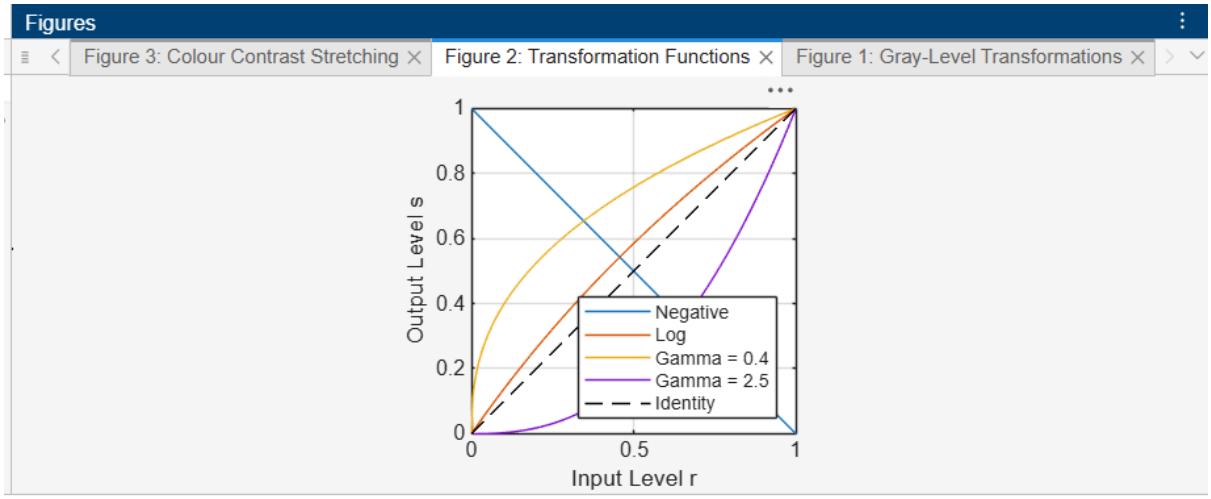


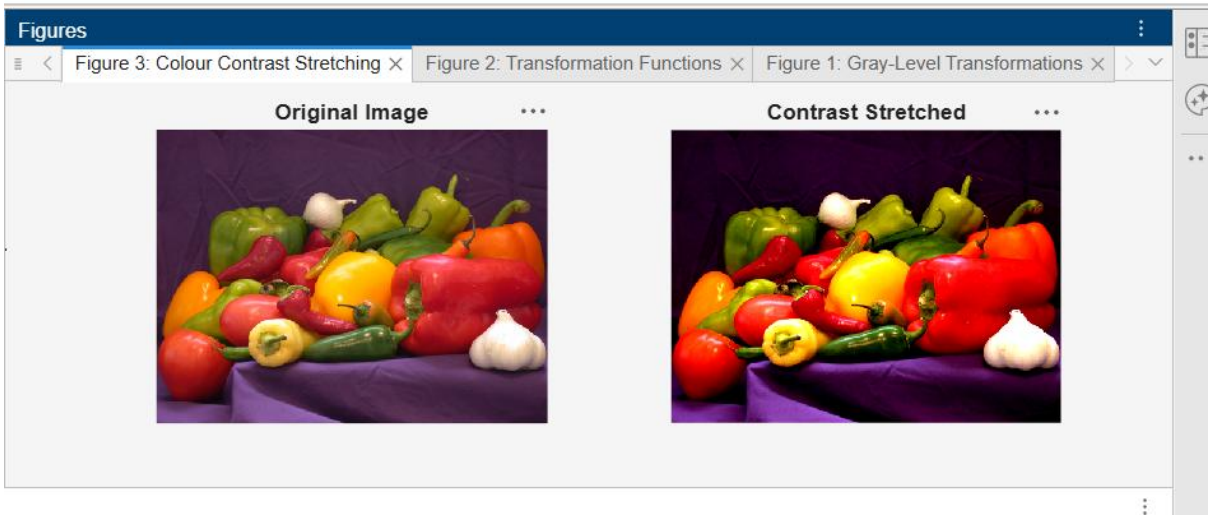
4. Procedure

1. Open MATLAB and set the Current Folder to your working directory.
2. Create a new script and save it as exp05.m.
3. Enter the program and run it by pressing F5.
4. Observe the original image and the negative, log, gamma and contrast-stretched images.
5. Examine the transformation curves displayed in the second figure.
6. Observe the contrast stretching applied to the colour image.
7. Examine the band-highlighted and binary-sliced images.
8. Experiment with gamma values of 0.2, 0.67 and 1.5.
9. Try intensity-slicing ranges of [0.3, 0.5] and [0.6, 0.9] and compare the results.
10. Save the output figures as PNG files and record the observations.

5. Expected Output







6. Observations

A. Gray-Level Transformations

Transformation	**Observation**
Original image	Low-contrast image
Negative	Gray levels are reversed
Log transformation	Dark regions become more visible

Gamma = 0.4	Image becomes brighter
Gamma = 2.5	Image becomes darker
Contrast stretching	Image contrast improves

8. Result

Image negative, log, power-law (gamma), contrast-stretching and intensity-slicing transformations were successfully applied using MATLAB. Their effects on image brightness, contrast and visibility of intensity details were studied.